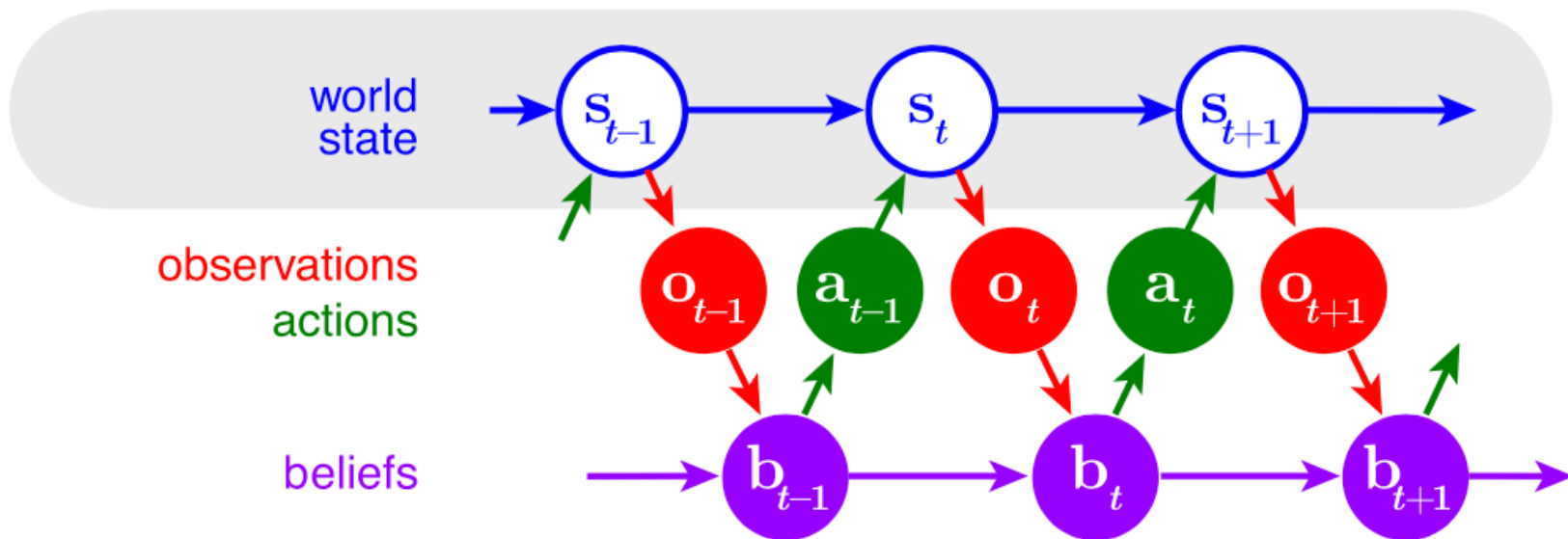


Estimating cost of attention using Inverse Rational Control

Inverse Rational Control (IRC)

- **Hypothesis:** Animal behavior not *optimal* but *rational*.
 - Rational w.r.t to its own internal costs and (possibly wrong) assumptions about the task.
- **Problem:** Can the observed behavior be used to estimate the internal costs and assumptions?

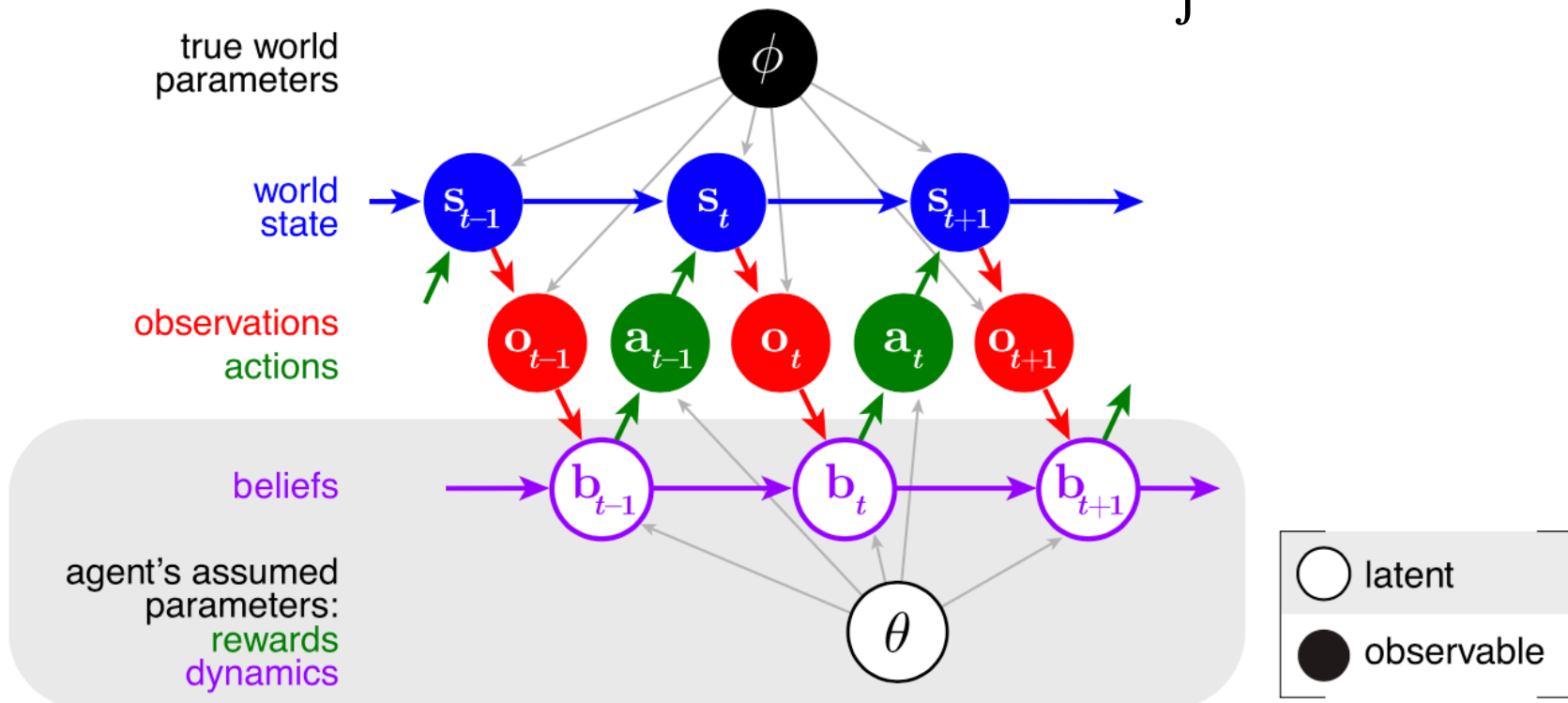
Partially Observed Markov Decision Process



IRC framework

What internal constraints and assumptions best explains the observed behavior?

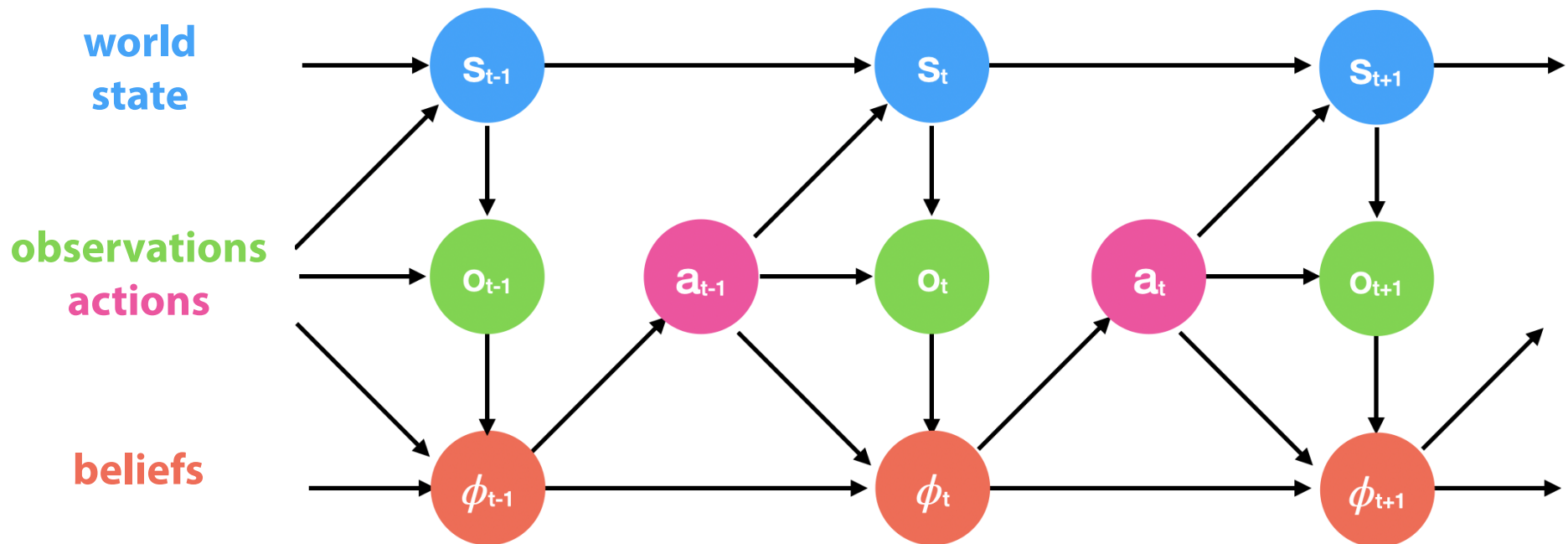
$$\mathcal{L}(\theta) = \log \int db_{1:T} p(b_{1:T}, o_{1:T}, a_{1:T}, s_{1:T} \mid \theta, \phi)$$



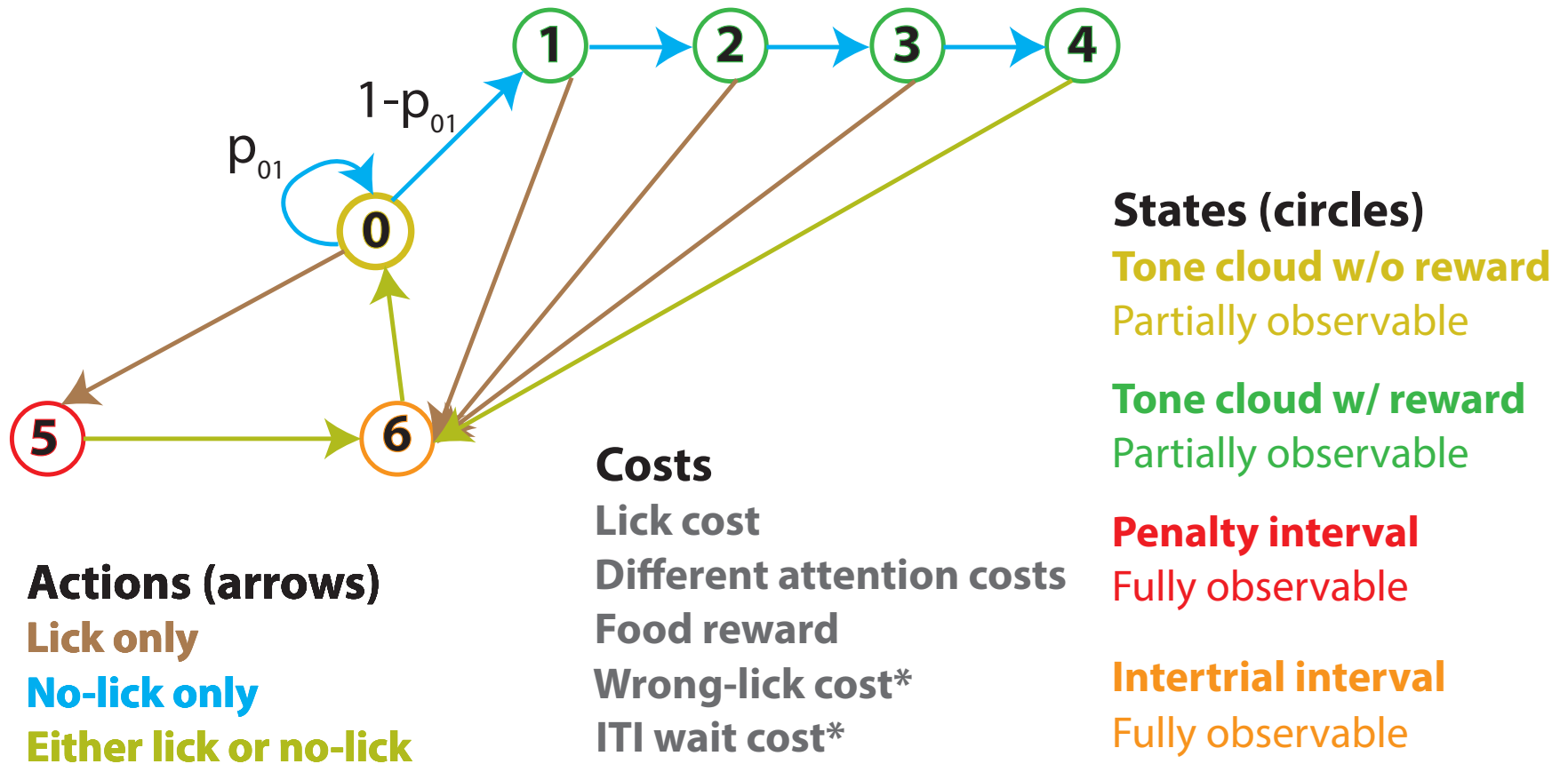
Auditory foraging toy problem

observations depend on attention level

actions = {attention level, lick choice}



Auditory foraging toy problem



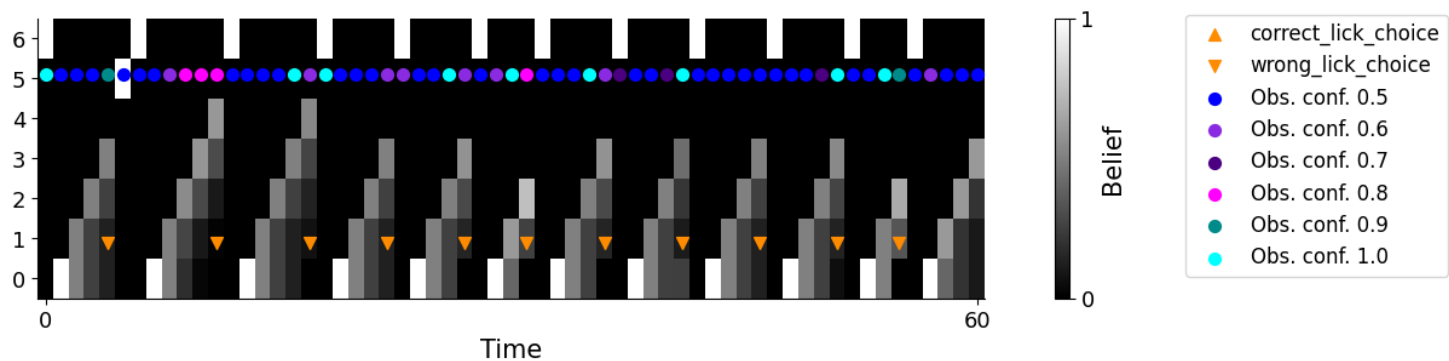
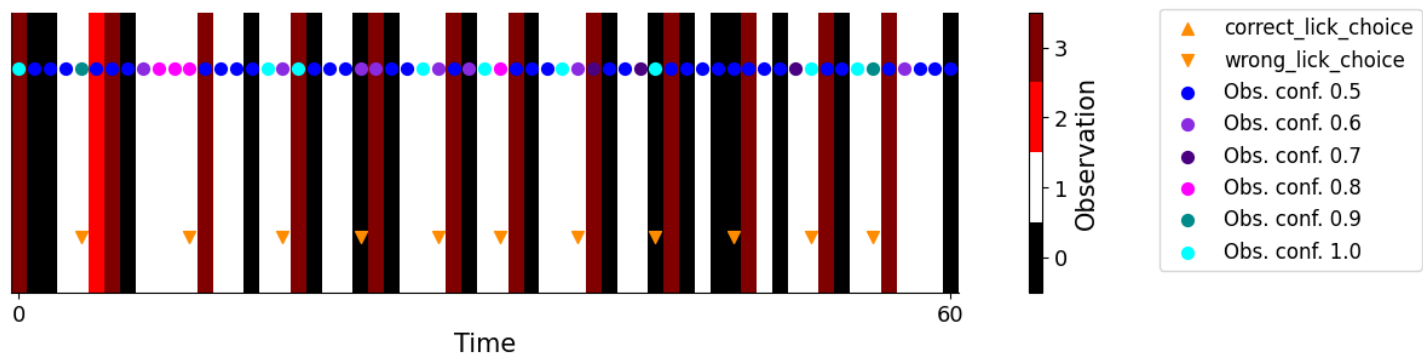
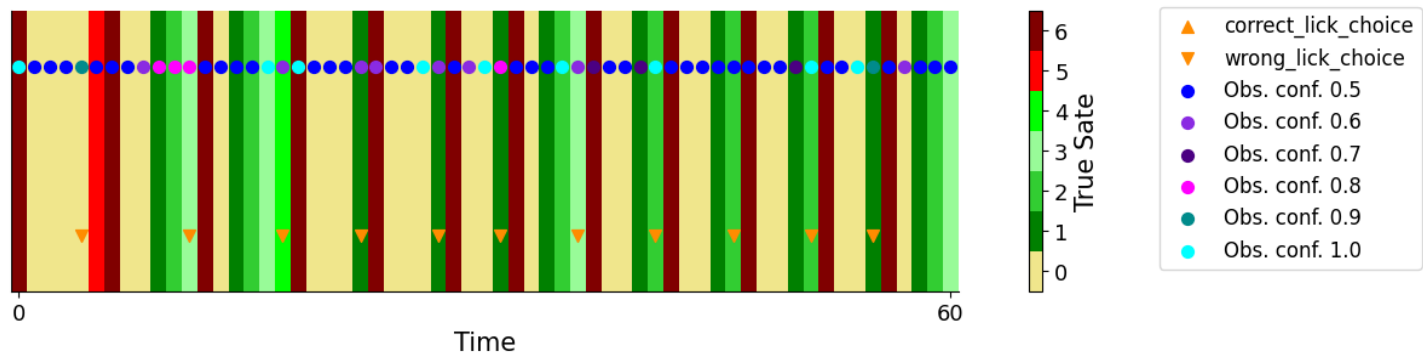
Auditory foraging toy problem

- **Observation dependence on attention level:**

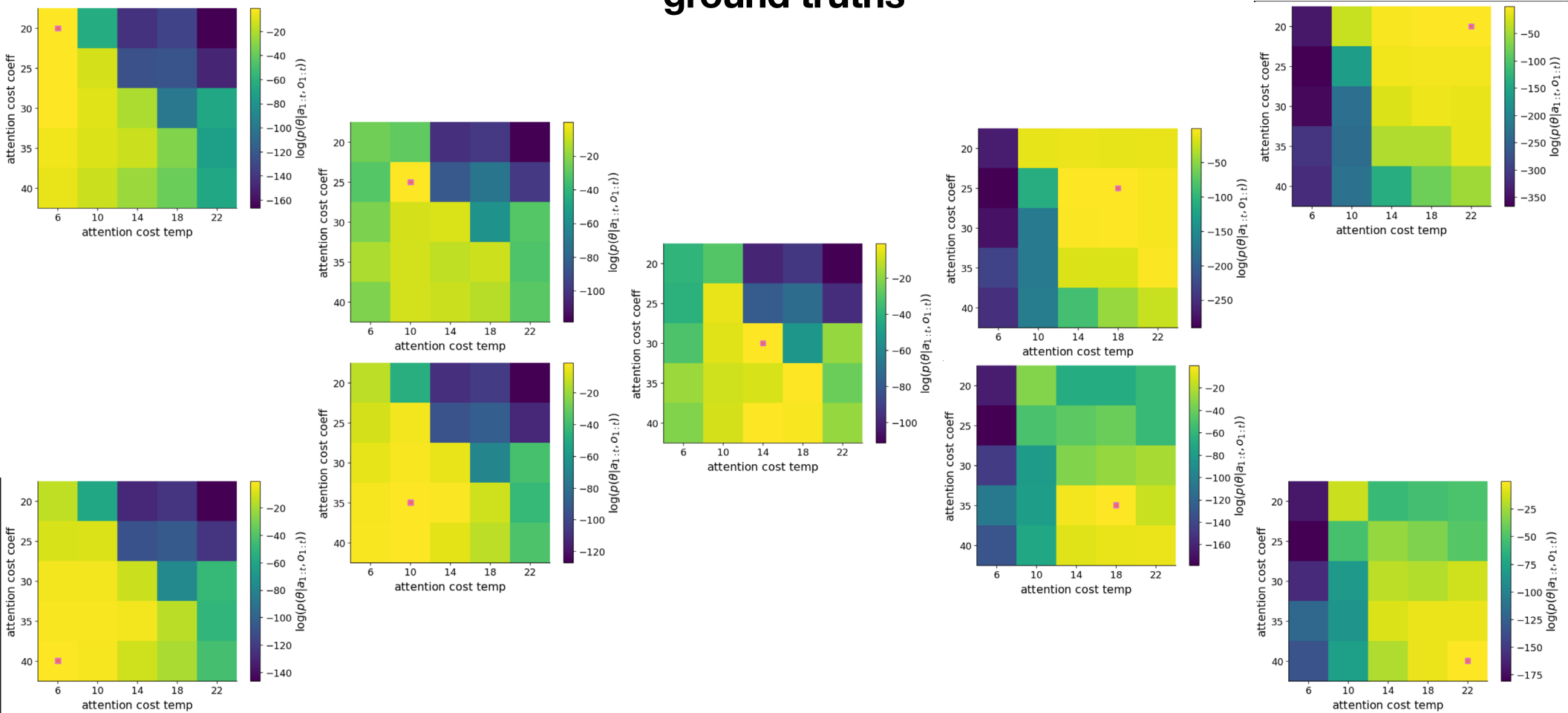
$$P(obs = food_{status} | obs_{conf}, food_{status}) = \text{Bernoulli}(obs_{conf})$$

- **Cost of attention:**

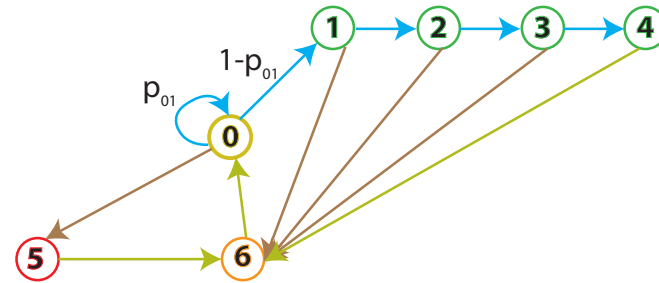
$$cost_{att}(obs_{conf}) = -coeff_{att} e^{\frac{obs_{conf}}{temp_{att}}}$$



Likelihoods for different ground truths



Current limitations:



- **Modeling penalty interval and inter-trial-interval as single nodes.**
 - Need to check if computation drastically increases otherwise.
- **Using discounted rewards as the objective function.**
 - Need to look into algorithms for average reward value as objective function.

Other challenges in applying to dataset?